

Dishank Oza

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SUMMARY

Software Engineer with 3+ years of professional experience in backend development, distributed systems, and performance optimization. Skilled in Java, Spring Boot, AWS, and databases, with a strong track record of designing scalable microservices, optimizing concurrency, and modernizing system architectures. Adept at building cloud-native applications and driving performance improvements.

EDUCATION

Masters in Computer Science - Santa Clara University | 4.0/4.0

Sept '24 - Mar '26

Bachelors in Computer Engineering - University of Mumbai | 3.5/4.0

Aug '17 - Jun '21

Relevant Coursework: Distributed Computing, Machine Learning, Object Oriented Analysis, Design and Programming, Database Systems, Data Structures, Advanced Algorithms.

TECHNICAL SKILLS

Programming & Database: Java, JavaScript, Typescript, Go, C/C++, HTML/CSS, PostgreSQL/MySQL, Cassandra, MongoDB, Neo4j, Redis

Libraries/Framework/Tools: React.js, Next.js, Vue Node.js, Tailwind, Spring Boot, Git, Jenkins, Jest, JIRA, Postman, Protocol Buffers, GraphQL

Cloud Technologies & Others: AWS, Firebase, Terraform, Docker, Kubernetes/EKS, Ansible, Grafana, Prometheus, Linux, Hadoop

Development Practices: Full-stack development, CI/CD, Automation, TDD, Design Patterns, Agile, SDLC, SOLID, and Code Reviews

Certifications & Publications: AWS Certified Developer Associate [\[Link\]](#), Scrum Alliance-Scrum Foundations, Discovering Backend Bottlenecks, Gesture-Based Interaction System [\[Link\]](#)

PROFESSIONAL EXPERIENCE

Optimizer Engineering Intern II, Teradata | SD, CA

Jun '25 - Sept '25

- Upgraded Teradata's database engine from **LLVM 16** → **20** by removing typed pointers, deprecated APIs, and redundant instruction casts, while updating codegen/JIT interfaces for compatibility and performance.
- Built IR-analysis tooling; validated on Teradata VM and AWS (1TB) with **~10% CPU-time** drop. Drivers: flatter GEPs, nuw, initializes(...), better inlining/aliasing, stronger loop vectorization.

Senior Software Engineer, Freespace | MUM, MH

May '23 - Aug '24

- Led a team of 4** in agile development, engineered business-critical features, and implemented CI/CD pipelines using Jenkins. Recognized as **Freespace Captain for leadership** and problem-solving.
- Architected and developed a microservice to streamline booking events using AWS DMS CDC, Kinesis, and EventBridge to distribute event generation from **30 to 1 events per user daily**, optimizing resource usage and system efficiency.
- Revamped API transaction handling by replacing serialized locks with **Redis distributed locks** in a Spring Boot application, reducing database load and improving concurrency, cutting application initial load time from **30 to 9s** during peak times.
- Pioneered scalable system architecture by designing and implementing the migration of booking APIs to an **event-driven** architecture with AWS SQS FIFO and Lambdas, **reducing server count from 12 to 8** during peak time.

Software Engineer, Freespace | MUM, MH

Jun '21 - May '23

- Led migration to single page application (SPA) with React routing, route-based and custom Webpack code splitting, HOCs, doubling first-load speed and improving repeat visits by **100%**.
- Developed a custom frontend component library created with React, TypeScript, and Storybook, delivering tailored branding styles for clients and boasting UI consistency and development efficiency.
- Designed and implemented a serverless notification system leveraging AWS Lambda, SQS, SNS, and SES for scalability, with multi-channel support (email, push, SMS) and a CloudFormation stack for automated infrastructure provisioning.
- Developed RESTful APIs using Spring Boot and GraphQL for an enterprise application serving 100,000+ users across 30+ global clients, managing up to **5,000** concurrent requests per minute.

PROJECTS

Stock Market Simulation Platform [\[github\]](#) | Java, Spring Boot

Aug '25

- Designed a Java (Spring Boot) stock-market microservices platform using Kafka with Redis/Postgres, instrumented with Micrometer/OpenTelemetry (metrics, tracing, Grafana dashboards) and containerized/deployed via Maven + Docker.

AmberDB – Distributed SQL Database [\[github\]](#) | Go, SQL

Apr '25

- Designed and built a scalable, fault-tolerant SQL database with Raft-based replication, MVCC using Hybrid Logical Clocks, dynamic sharding, and 2PC for atomic cross-range transactions.

Spacelink [\[github\]](#) | Next.js, React

Nov '24

- Developed a 2D virtual space platform for real-time avatar interactions, with WebSockets and Protobuf for efficient data transfer. Integrated video calls via WebRTC and managed data with PostgreSQL, using TypeScript, Prisma, and Next.js.

ACHIEVEMENTS, AWARDS, AND LEADERSHIP

- Awarded Freespace Captain for successfully leading the team, utilizing exceptional problem-solving skills, and ensuring timely delivery of project deliverables
- 3rd in Smart India Hackathon qualifier 2020 for license plate recognition and automated parking system using Raspberry Pi.
- Member of Team Vaayushastra, ranked 9th globally and 2nd in the Asia-Pacific region in the SAE Aero Design Competition 2020.